

2026 INTERVIEW GUIDE

30

LLM Fundamentals Interview Questions

The questions every AI interview starts with.
Answered in plain words, with real code.

Tokens

How LLMs Work

Prompting

RAG & Agents

Production



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READ THIS FIRST

The screening round doesn't test genius. It tests **foundations**.

Thirty questions come up before any system design. Answer them simply and correctly, and you earn the harder rounds. Fumble them, and the interview ends politely.



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HOW SCREENING ROUNDS RUN IN 2026

5 areas. 30 questions.

1 Tokens & Context Q1-6

2 How LLMs Work Q7-12

3 Prompting & Control Q13-18

4 RAG, Agents & Tools Q19-24

5 Production Basics Q25-30

Every answer here is short, plain, and sayable out loud.
That's the whole trick.



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Q1 What is a token?

- ▶ A piece of a word. Models read and write tokens, not letters. One token is roughly **three-quarters of a word** in English.
- ▶ Why it matters: you pay per token. Limits are in tokens. Speed is tokens per second.

```
import tiktoken
enc = tiktoken.get_encoding("o200k_base")
len(enc.encode(text)) # count before you pay
```

Q2 Why can't the model count letters in "strawberry"?

- ▶ Because it never sees letters. It sees **tokens**, and "strawberry" may be one or two chunks.
- ▶ This answer shows real understanding. Tokenization explains many odd failures: spelling, rhymes, arithmetic on digits.



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Q3 What is a context window?

- ▶ The model's **working memory** for one request. Everything must fit: instructions, documents, chat history, and the answer.
- ▶ It is not storage. Close the conversation, and it is gone. Persistence is your job, not the model's.

Q4 Big context windows exist now. Problem solved?

- ▶ No. Cost grows with **every token you send**. And models still miss facts buried in the middle.
- ▶ The senior line: a big window is a bigger desk, **not a better librarian**. You still retrieve first.

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Q5 What does temperature actually do?

- ▶ It controls how **random** the word choices are. Low means predictable. High means adventurous.
- ▶ Rule: near **0 to 0.3 for factual work**. Higher only for brainstorming and variety.

```
llm.invoke(prompt, temperature=0.2) # grounded answers
```

Q6 What's the difference between max_tokens and the context window?

- ▶ The context window is the **total budget**, input plus output. max_tokens caps only the **answer length**.
- ▶ Set max_tokens too high, and long inputs leave no room to answer. Interviews love this small trap.

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Q7 How does an LLM actually work, in one minute?

- ▶ It predicts the **next token**, again and again. That's the whole mechanical loop.
- ▶ The magic is scale: trained on enormous text, prediction starts to **look like reasoning**. Keep the answer this simple. Simplicity signals mastery.

Q8 What does "8B parameters" mean?

- ▶ Eight billion learned numbers, the model's **tuned knobs**. More parameters mean more capacity, cost, and latency.
- ▶ Bigger is not always better. A small model with retrieval often **beats a big one without it**.

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Q9

Explain attention in plain words.

- ▶ For each new token, the model asks: **which earlier words matter right now?** Attention scores every one, then focuses.
- ▶ In "the cat sat because it was tired", attention links **it** to **cat**. That linking is the breakthrough.

Q10

Why do models hallucinate?

- ▶ Because they always answer with the **most plausible** next tokens. Plausible and true are different things.
- ▶ It's not lying. It's confident autocomplete. Fixes: **grounding, citations, and an allowed "I don't know"**.



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Q11 What is a knowledge cutoff?

- ▶ The date training data ends. The model knows **nothing newer**, unless you provide it.
- ▶ That's why fresh facts come from **retrieval or tools**, never from the model's memory.

Q12 Same prompt, different answers. Why?

- ▶ Sampling is **random on purpose**. Temperature above zero means varied word picks each run.
- ▶ For repeatable behavior: temperature 0, versioned prompts, and **evals** to catch drift anyway.

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Q13 System prompt vs user message. What's the difference?

- ▶ The system prompt is the **standing contract**: role, scope, rules. User messages are the conversation.
- ▶ Separation matters. Rules in the system prompt **survive the whole chat** and resist override attempts.

```
messages=[{"role": "system", "content": rules},  
           {"role": "user", "content": question}]
```

Q14 What is few-shot prompting?

- ▶ Showing **two or three worked examples** instead of longer instructions. The model copies the pattern.
- ▶ Three good examples beat three hundred words of rules. Especially for **format and labeling**.



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Q15 How do you get reliable JSON out of a model?

- ▶ Never by asking politely. Use **schema-enforced structured output**, then validate, then retry once.
- ▶ "Respond in JSON" fails a few percent of the time. At scale, that's an **outage**.

```
response_format={"type": "json_schema", ...}  
Invoice.model_validate_json(reply) # retry on failure
```

Q16 What are reasoning models, and when do you use them?

- ▶ Models that **think before answering**, internally, step by step. You set an effort level.
- ▶ Use them for multi-step logic and planning. Skip them for lookups: **slower and costlier for nothing**.

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Q17 The model keeps ignoring an instruction. What do you check?

- ▶ Three usual suspects. The rule is **buried mid-context**. It's phrased as a don't. Or it conflicts with another rule.
- ▶ Fixes: move it to the system prompt, make it **positive and specific**, and validate in code.

Q18 How do you pick a model size for a task?

- ▶ Start with the **smallest model that passes your eval**. Escalate only proven gaps.
- ▶ Routing routine work to small models is the **single biggest cost lever** in production.



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Q19 Explain RAG in three sentences.

- ▶ Search your documents for relevant passages first. Give them to the model. It answers **from that evidence**, with citations.
- ▶ Why it wins: fresh knowledge without retraining, and answers you can **verify**.

Q20 Explain fine-tuning in three sentences.

- ▶ Additional training on your examples. It changes **behavior**: tone, format, habits.
- ▶ It is a poor way to add facts. The line: **fine-tune the interface, retrieve the content**.

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Q21 What is an embedding?

- ▶ Text turned into a **list of numbers** that captures meaning. Similar meanings sit close together.
- ▶ That's what makes semantic search work: "refund policy" finds **"getting my money back"**.

```
emb = model.encode(chunks, normalize_embeddings=True)
```

Q22 What makes something an agent?

- ▶ The model **decides the path**: which tools, what order, when it's done. A loop, not a single call.
- ▶ The model never executes anything. It asks. **Your code executes**. That boundary is the security story.

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Q23 What is MCP?

- ▶ An open standard connecting AI apps to tools and data. Build the integration **once**, every model can use it.
- ▶ The USB-C analogy lands: one port, many devices. In 2026 it's the **default enterprise wiring**.

Q24 What guardrails would you put around an LLM feature?

- ▶ Input checks for injection and PII. Output validation against a schema. **Approval gates on risky actions**.
- ▶ The prompt is never the security boundary. **Deterministic code** is.

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Q25 Where does the money go in an LLM feature?

- ▶ **Input tokens**, usually. Fat contexts on every call dwarf the short answers coming back.
- ▶ Levers: trim context, cache the stable prefix, route to **smaller models**. Measure cost per successful task.

```
cost = tok_in/1e6*p_in + tok_out/1e6*p_out  # per call
```

Q26 What latency number matters most?

- ▶ **Time to first token**. Users forgive a slow finish, never a silent screen.
- ▶ So stream every answer, and keep the first word **under a second**.

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Q27 What is prompt caching?

- ▶ Providers reuse the **unchanged start** of your prompt across calls, at a fraction of the price.
- ▶ Design for it: stable system prompt and examples **first**, changing content last.

Q28 How do you know your LLM feature actually works?

- ▶ **Evals.** Fifty real questions with known good answers, run on every change.
- ▶ "We tested it and it looks good" is the answer that **ends interviews**. A golden set is the answer that starts offers.

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Q29

The provider ships a new model. What do you do?

- ▶ Treat it as a **deploy, not an upgrade**. Run the eval suite, canary a traffic slice, keep rollback ready.
- ▶ Model swaps silently change behavior. Teams that skip the eval find out **from users**.

Q30

Is the bigger model always the better choice?

- ▶ No. The better choice **passes your eval at the lowest cost and latency**.
- ▶ Saying that sentence, calmly, is the whole point of this deck. Foundations plus judgment is what screening rounds hire.

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Preparing for AI interviews?



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One deep guide every morning. One **15-second answer** every afternoon at 3.